

## Introduction

Productivity and time play an extremely important role in both students and professors' daily lives especially in this AI age where technology and engineering keep improving and new trends pop up every day. For example, lecturers and professors should not spare time for monitoring the exam rooms for students. They should have "Virtual Proctor Agent" to help with students which will help them save time and manpower instead of being in the rooms physically.

| Agent type            | Performance measure                                      | Environment                   | Actuator                                           | Sensors                                                                               |
|-----------------------|----------------------------------------------------------|-------------------------------|----------------------------------------------------|---------------------------------------------------------------------------------------|
| Virtual Proctor Agent | Accuracy, Real-time response, user-experience, fairness, | Physical, digital, behavioral | Alerts, Automated actions, Data Recording, Reports | Web cams, Stand-up cameras, Microphones, Screen Monitoring Software, Network monitors |

Figure 1: PEAS description of the task environment for Virtual Proctor Agent

## Performance Measure

**Accuracy:** To ensure that the agent correctly identify cheating behaviours without false positives (flagging innocent behaviours as cheating) or false negatives (failing to detect actual cheating).

**Real-Time Response:** The speed at which the agent detects and alerts for suspicious behaviours.

**User Experience:** Minimizing disruption to students while ensuring effective monitoring.

**Fairness:** The system inside the agent must be unbiased, ensuring all students are monitored equally regardless of background.

## Environment

**Physical Environment:** The student's surroundings during an exam, including room layout, lighting conditions, and the presence of unauthorized materials or individuals.

**Digital Environment:** The student's computer or device, including screen activity, open applications, internet usage, and any attempts to access unauthorized resources.

**Behavioural Environment:** Student behaviours during the exam, such as eye movements, facial expressions, body posture, and any signs of distraction or communication with others.

### **Actuator**

**Alerts:** The agent must send notifications to proctors or exam administrators when suspicious activity is detected.

**Automated Actions:** To ensure that actions like pausing the exam, capturing screenshots or video clips, logging the student out, or locking the screen should be automated.

**Data Recording:** The agent stores logs, screen recordings, and video/audio clips of suspicious activity for post-exam review.

**Reports:** To make the agent generate summary reports of flagged incidents for further investigation.

### **Sensors**

**Webcams:** Monitoring the student's physical environment, including face, eyes, and body movements.

**Microphones:** Detecting unusual sounds, such as other people talking, or verbal communication with unauthorized parties.

**Screen Monitoring Software:** Tracking the student's screen activity, including active windows, copy-pasting behaviours, and attempts to access unauthorized websites or applications.

**Network Monitors:** Detecting suspicious network activity, like the use of VPNs, or access to external resources that are not permitted during the exam.

| Task environment       | Observable | Agent | Deterministic | Episodic   | Static  | Discrete   |
|------------------------|------------|-------|---------------|------------|---------|------------|
| Rooms, Digital devices | Partially  | Multi | Stochastic    | Sequential | Dynamic | Continuous |

Figure 2: The task environment and the characteristics of Virtual Proctor Agent

#### Task Environment:

**Partially observable:** The Virtual Proctor Agent cannot see everything that is happening in the environment. It relies on sensors like webcams, microphones, and screen monitoring, which might not capture all cheating behaviours or external factors.

**Multi-Agent:** The environment involves multiple students (potentially many agents), each of whom the system must monitor. The students might interact with the environment in ways that could affect the system's behaviours.

**Stochastic:** The environment has elements of unpredictability. For example, students may engage in unforeseen cheating methods, or external factors like lighting changes may affect the agent's ability to monitor effectively.

**Sequential:** The agent must continuously monitor students over time. Decisions and actions are interdependent, as initial behaviours may influence subsequent actions, and past context is important for making decisions (e.g., a student looking away from the screen may only be suspicious if it happens repeatedly).

**Dynamic:** The environment can change during monitoring. For instance, students can move, change their behaviours, or interact with unauthorized materials over time, requiring the agent to adapt and respond.

**Continuous:** Monitoring involves continuous streams of data, such as video feeds and network activity. The agent must make decisions in real-time based on this continuous input.